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Can LLMs predict the success of Turkish TV series from their first episodes?

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ABSTRACT

Turkey is the third largest exporter of TV series worldwide. However, half of these series are cancelled early leading to economic and social consequences. In this study, we explore whether the success of these series can be predicted from the scripts of their first episodes using LLMs. We built a dataset of first-episode scripts from recently aired Turkish series and trained LLM-based models on it. The main challenge we faced is that these scripts are very long, making them unsuitable for standard BERT models. This led to one of the key contributions of our study, as there is currently no research that specifically focuses on handling long Turkish texts. We pretrained a BigBird model from scratch for Turkish and fine-tuned it for our task. We also developed a Hierarchical Attention Network (HAN) model capable of processing long Turkish texts. While predicting the exact number of episodes is difficult, both HAN and BigBird achieve strong performance in binary classification setup, distinguishing successful series from unsuccessful ones. Additionally, we investigate whether audience preferences in Turkey have changed over time by testing our models on some iconic older Turkish series to see if they would still be classified as successful by today's standards.

1. Introduction

Large Language Models (LLMs) are advanced deep learning models trained on massive text data to understand and generate human language. With the latest breakthroughs in the transformers and GPU technology, LLMs have achieved outstanding performance than ever [1,2]. In particular, GPT-based models, a special type of LLMs that focus on text generation, have ushered in a new era. This has made Artificial Intelligence (AI) accessible and practical for nearly everyone worldwide. In fact, these models have revealed the true potential of AI and shown that some of the great expectations about AI are finally starting to come true.

LLMs have therefore found successful applications in areas as diverse as law, cybersecurity, CRM (customer relationship management) and language translation [2,3]. Each successful application has naturally opened the door to new applications and media and entertainment has been one of them [4]. LLMs — and AI more broadly — have proven useful in recommending TV shows, movies and music to targeted audiences [5]. Beyond recommendations, the media and entertainment industry uses LLMs primarily for content creation, particularly through Generative AI (Gen-AI) [4,6]. Dramatron from DeepMind is a good example of this. It helps industry professionals write scripts by generating coherent narratives, characters and dialogue, and can thus be used as a co-author for screenplays. In addition, Google recently introduced Flow, an AI filmmaking tool capable of creating realistic cinematic videos that can reflect physical truths, while the user only needs to define a subject or a scene.

Regardless of whether an AI or a human creates a screenplay, it is crucial to predict how the audience will respond before producers invest millions of dollars in a project. In this study, we argue that LLMs can help us predict the success of entertainment content based on its text-based features. More specifically, we are interested in predicting the lifespan of Turkish TV series based on the script of the first episode, using different NLP models including LLMs based on Bidirectional Encoder Representations from Transformers (BERT) [7].

We focus specifically on Turkish TV series as they have a significant and growing global impact. According to Statista, The Turkish television and video market is estimated to reach US$2.38bn in 2025.[1] In fact, Turkish dramas have become major cultural export and are purchased by more than 170 countries, reaching an estimated 800 million viewers worldwide. The Economist claims that Turkey is the third largest exporter of television content after the US and the UK.[2] However, since both the US and the UK primarily produce content in English — and much of the existing literature on LLM applications also focuses on English - it becomes both more interesting and necessary to study Turkish TV series produced in Turkish, categorized as a low-resource language [8,9]. In fact, the application of LLMs to

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[1] https://www.statista.com/outlook/amo/media/tv-video/turkey.

[2] https://www.economist.com/culture/2024/02/15/the-third-largest-exporter-of-television-is-not-who-you-might-expect.

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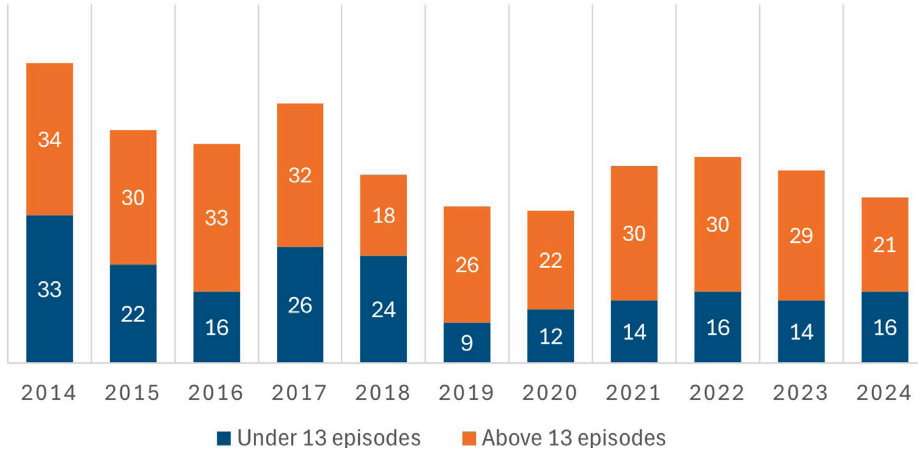


Fig. 1. The total number of TV series aired in the major TV channels of Turkey in between 2014–2024, distinguished between those that aired under 13 episodes and those that above 13 episodes. Note that the numbers given in this figure are based on data extracted from Wikipedia entries on Turkish television series [13].

Turkish-language content is still relatively under-researched compared to English. For example, in the case of the widely known LLM GPT-3, approximately 93% of the training tokens were in English, while only about 7% represented all other languages combined, including Turkish [10].

Each year, about 60 to 70 TV series are produced in Turkey by major media companies for a public audience. While digital platforms are becoming popular, most of these series are still broadcast on the country's main stream television channels [11]. Due to their failure to achieve sufficient popularity in a short period of time — typically measured by TV ratings — nearly half of these series are cancelled after just a few episodes. Specifically, it is common for producers and TV channels to sign an agreement for 13 episodes Therefore, series that air fewer than 13 episodes are typically considered unsuccessful [12]. Fig. 1 presents the total number of TV series broadcast on major Turkish TV channels over the past ten years, distinguishing between those that aired fewer than 13 episodes and those that exceeded 13 episodes. Note that nearly half of the TV series each year were canceled early, before reaching the 13-episode threshold.

It is clear that the cancellation of a TV series has a number of economic and social consequences. These consequences include but not limited to sunk costs for producers, reduced advertising revenue for TV channels, job losses for cast, crew and support staff, a loss of export revenue and gaps in broadcasting schedules that can lead to lower ratings [14]. It is therefore of paramount importance to forecast the success of a series before shooting begins. This task is typically carried out by human reviewers, such as producers or content editors at broadcasting companies. However, such forecasting is often influenced by external factors, including reviewer biases, internal politics within networks, and even commercial pressures from sponsors or advertisers. In this study, we argue that more objective and reliable predictions can be made using AI models trained on data from previously aired TV series.

We collected scripts of around 200 TV series that aired on Turkey's major mainstream channels — Star, Show TV, Now (formerly Fox), ATV, Kanal D and TV 8 — over the past five years, from 2020 to the present.[3] This focus is driven by the rapidly evolving nature of audience preferences and industry dynamics [15]. By analyzing recent series, we try to capture current trends and viewer expectations and ensure that our forecasting models are in line with today's television landscape. Since each episode of a Turkish TV series runs for nearly two hours, their scripts can be up to 80–90 pages long. This makes it challenging to use them directly with standard BERT models, which are limited to processing a maximum of 512 tokens. To deal with this, we extended the BERT architecture to handle longer inputs. Specifically, we developed a BigBird [16] model for Turkish, a sparse variant of BERT designed to work with tokens up to 4096 tokens. To best of our knowledge, this is the first BERT-based method that can handle long Turkish texts. In addition, we developed a Hierarchical Attention Network (HAN) [17] model for Turkish which is also capable of handling Turkish long texts. We compared the performance of these BERT-based models with a more traditional approach to text classification and regression, which involves using TF-IDF vectorization followed by classification with either a random forest or logistic regression model to understand if LLMs are truly necessary for the tasks we are interested in.

Using our BERT-based large language models (LLMs), and ignoring external factors such as cast, broadcasting channel, and airing season, we aim to address the following research question:

Can the success of Turkish TV series be predicted solely from the scripts of their first episodes, using LLMs trained on data from previously aired shows?

In addition to the main question above, we also seek to explore whether audience preferences have shifted over time. Specifically, we aim to examine whether our models, trained on recent TV series thus reflecting current trends in Turkish television, would still consider certain older series, which were highly popular in the past, to be successful.

Answering the questions above led us to make two original contributions. First, we worked with long Turkish texts, something that has not been explored much before. Specifically, we pretrained a BigBird model for the Turkish language using the Turkish Wikipedia corpus from scratch and shared it on HuggingFace.[4] We then added a classification/regression head to fine-tune it on our TV series dataset. That said, this model can also be fine-tuned for other NLP tasks involving long Turkish texts, such as Named Entity Recognition, Sentiment Analysis, or Question Answering. Second, this is the first study — at least to our knowledge — that tries to predict the lifespan of Turkish TV series before they even air, which could have a meaningful economic impact on the media industry.

In the rest of the paper, we first review related work that aligns with our study, and then introduce the models we developed. After that, we present and discuss the results of our experiments in detail. We proceed with the limitations of this study. Finally, we conclude the paper with a summary of our findings and potential directions for future research.

[3] We did not include TV series from TRT 1, Turkey's main state broadcaster, in this study, as the lifespan of its series does not always depend on audience appreciation and may instead be influenced by government policies.

[4] https://huggingface.co/FiratIsmailoglu/bigbird-turkish-pretrained

2. Related works

We divide this section into two parts. In the first part, we review studies related to predicting the success of TV series and other entertainment products, such as movies and songs. In the second part, we focus on more technical research that addresses the challenges of processing long documents in Natural Language Processing (NLP), since the scripts of TV series we are interested in are considered long documents, i.e. consisting of more than 512 tokens.

2.1. Predicting the success of entertainment products

It is well-established that in a wide range of prediction tasks, from image classification to natural language understanding AI models can surpass human performance [18]. Because of this, they have been widely adopted across various industries including entertainment. This has led to several successful applications, like predicting hit songs or forecasting box office performance, to name but a few. Below, we provide a brief summary of these applications.

ScriptBook is an AI startup that aims to predict a movie's box office performance at pre-production and pre-release stages by analyzing its script.[5] Their models were trained on thousands of past movie scripts and can predict who is likely to enjoy a film based on demographic factors like age group and gender. ScriptBook also offers its premium users a feature called *Territoriality Forecast* which gives an insight into which countries will be interested in the movie to be released. On the academic side, one of the earlier notable studies is [19], which focused on predicting the ratings of five popular sitcoms (e.g., The Office, The Big Bang Theory, etc.) using features like the title, directors, writers, original air date, and the number of lines spoken by each character extracted from sitcom transcripts. The authors achieved high accuracy using standard machine learning algorithms such as decision trees and neural networks. However, the models were tailored to each specific sitcom, since the extracted features were unique to them. As a result, the approach lacks generalization and cannot be easily applied to other TV series.

A more recent study is [20], which focuses on predicting the ratings of Turkish TV series using attributes like the popularity of previous episodes, genre, originality, and search trends. However, this approach relies on data that can only be collected after the show has aired, such as past ratings, viewer interest, and social media interactions. Because of that, their prediction model is not suitable for use at the pre-production stage, before any investment is made in the series. Another notable study in this area is [15]. The authors collect sequences of daily tweets for 75 popular TV shows in the U.S., which are used as input to a Long Short-Term Memory (LSTM) model designed to capture temporal patterns in social media activity. Based on these patterns, they predict the TV rating scores of upcoming episodes. Similar to [20], this approach depends on data that can only be gathered after some episodes have already aired, so it does not directly compete with our study. Still, the idea behind their method is not limited to TV series, instead, it can be applied to any kind of TV show in the market.

Obviously, the use of the predictive power of AI in the entertainment industry goes beyond TV shows and movies. A great example is hit song prediction. One interesting study in this area is [21], where researchers use listeners' neurophysiological responses collected via wearable neural devices as features. After each song, participants answered questions like how much they liked it and whether they would recommend it to friends, which were used as labels. The researchers then trained standard machine learning models like k-NN and SVM on this data and achieved strong results. Their models were able to predict whether a new song would be a hit or not, based on neurophysiologic responses from listeners who had not heard the song before. Speaking of hit song prediction, [22] trained a large language model (LLM) using the lyrics of over ten thousand pop songs released in the UK. Their goal was to predict the Spotify popularity score on day 15 after the release of a song. The study showed that lyrical features play a significant role in predicting music popularity and contribute notably to achieving high accuracy.

In printed media, a noteworthy study is [23], where the researchers try to predict book sales before publication. Instead of using the book's actual content, they represent each book through features related to the author (like visibility and past sales), the book itself (such as genre and topic), and the publisher (like its reputation). While this method works well for known authors and publishers, it faces the cold start problem, as it is nearly impossible to predict the success of a book from a brand-new author or publisher, since there is no prior data available at the time of release. Finally, a master's thesis explored predicting the success of video games, which was measured by the average number of players on the Steam platform, using pre-release features such as the number of characters, storage requirements, genres, and the level of graphical content, through machine learning models [24].

2.2. Handling long texts with LLMs

Large Language Models (LLMs) based on the Transformer architecture have led to significant advances across a wide range of NLP tasks including language inference, text classification/regression and machine translation to name a few. Among these models, BERT (Bidirectional Encoder Representations from Transformers) emerges as a key architecture that has shaped much of the recent progress in NLP [7]. Concretely, BERT introduced the idea of deep bidirectional contextual encoding through masked language modeling (MLM), which allows it to learn deep bidirectional representations, meaning it can take into account both the words before and after each token in a sentence. This approach has helped BERT reach state-of-the-art performance on several major benchmarks, like GLUE and SQuAD. Thanks to its flexibility, BERT has become the preferred model for a variety of downstream tasks, such as document classification, question answering and named entity recognition [25].

A common criticism of BERT and many of its variants is that they cannot handle long sequences, i.e., anything longer than 512 tokens, which reduces its direct applicability to tasks that require understanding longer contexts [16,26,27]. The main reason is that in a standard Transformer, each token attends to every other token in the sequence, which causes the time and memory usage to grow quadratically. For this reason, most pre-trained BERT models limit the input length to 512 tokens. Early studies tried to deal with long texts by simply truncating them, taking the first 512 tokens, or the same amount tokens from the middle or the end. But this approach often missed important information, so it was quickly dropped. Later on, more advanced techniques were introduced to handle long texts in NLP. These methods generally fall into two main categories: hierarchical approaches and models that use sparse attention mechanisms.

Hierarchical models handle long texts by dividing them into smaller chunks, each with a maximum of 512 tokens, and then process each chunk separately using a pretrained BERT model. Once each chunk is encoded, the outputs are combined to form a final representation of the full (long) document, which is then passed to a classification head. A well-known example of this approach is [28], where two versions of this type were introduced: Recurrence over BERT (RoBERT) and Transformer over BERT (ToBERT). In RoBERT, each chunk is encoded using BERT, and the [CLS] token output (from the last transformer block) is obtained. These [CLS] embeddings are stacked into a sequence, which is then fed into an LSTM. The LSTM's output serves as the document-level embedding. ToBERT works similarly, but instead of an LSTM, it uses a small Transformer to process the sequence of [CLS] outputs.

Hierarchical Attention Networks (HANs) were first introduced by Yang et al. in 2016 for document classification tasks [17]. Later,

[5] https://www.scriptbook.io/.

Chalkidis et al. [29] demonstrated that HANs can also be effectively applied to long documents. The core idea behind HANs is that documents are made up of sentences, and sentences are composed of words. As a result, HANs first break a document into sentences, and then tokenize each sentence into words. These words are vectorized using a pretrained Word2Vec model and passed through a word-level attention mechanism to generate sentence embeddings. The sentence embeddings are then fed into a sentence-level attention layer to produce a document embedding. Finally, a classification head, typically a fully connected layer, is applied to this document embedding to make the final prediction. As we use HANs for our experiments, we will explain them in more detail in the next section.

As for the models designed to handle long documents using a sparse attention mechanism two architectures, namely BigBird [16] and Longformer [27], stand out. Unlike standard BERT, which scales quadratically with input length, these models scale linearly, allowing them to handle much longer sequences, up to 4096 tokens. Both models employ hybrid attention mechanisms that combine local and global attention. For local attention, each token attends to a fixed number of neighboring tokens, capturing nearby context efficiently. For global attention, some certain tokens (e.g., [CLS]) are allowed to attend to all other tokens in the sequence, enabling the model to capture important information spread across the entire document. BigBird further incorporates random attention allowing some random tokens attend to other tokes, like [CLS] token does, which eventually leading to approximate full attention mechanism used in standard BERT models. As a result, BigBird improves its ability to model long-range dependencies. Because of these advantages, we chose BigBird in our experiments.

3. Prediction models

In this section, we explain in detail the models we used to predict the success of Turkish TV series based on the scripts of their first episodes.

3.1. TF-IDF based models

Before the rise of large language models (LLMs), the standard approach for many NLP tasks was to represent texts using the TF-IDF method. In this method, documents are first converted into vectors using the classic bag-of-words technique. Then, each word's count (term frequency—TF) is multiplied by its inverse document frequency (IDF), which is calculated by dividing the total number of documents by the number of documents that contain that word. This weighting reduces the impact of very common words (like articles) by assigning them lower scores, while giving higher scores to rarer words that are more likely to carry meaningful information.

To establish a baseline and explore whether LLMs are truly necessary for predicting TV series success, we first applied the TF-IDF vectorizer from the scikit-learn library.[6] Using the resulting text representations, we built two traditional machine learning classification models: one using Random Forest and the other using Logistic Regression, both also provided by scikit-learn. We refer to these baseline models as TF-IDF-RF and TF-IDF-LR, respectively. We also used TF-IDF features for linear regression models to predict the total number of episodes a series would have. Specifically, we applied Lasso and Ridge regressions, which are linear regression models with L1 and L2 regularization respectively, using the scikit-learn library.

[6] https://scikit-learn.org.

3.2. Truncated BERT model

As mentioned earlier, standard BERT models cannot process texts longer than 512 tokens [7]. A simple but commonly used workaround for this limitation is to truncate long texts by taking only the first 512 tokens. To evaluate whether more complex BERT-based models designed for long sequences are truly necessary for our task, we also implemented this truncation strategy, resulting in a model we refer to as Truncated-BERT. Specifically, this model is built on top of a pretrained Turkish BERT model, dbmdz,[7] which was trained on the Turkish OSCAR corpus and a recent Turkish Wikipedia dump—amounting to a total of approximately 44 billion tokens. To fine-tune this model for our prediction task, we added the following classification head:

The [CLS] output in Fig. 2 represents the embedding of the [CLS] token produced by the pretrained BERT model. In our experiments, we also evaluated variants of the truncated BERT model using the last 512 tokens and a middle segment of 512 tokens. However, these alternatives did not result in any notable improvement in model performance compared to using the first 512 tokens, we therefore only report the results of the model using the first 512 tokens.

3.3. Hierarchical Attention Network model (HAN)

We built a Hierarchical Attention Network (HAN) model for predicting the success of Turkish TV dramas, following the original design proposed in [17]. Concretely, after splitting each script (i.e., document) into sentences and tokenizing those sentences into words, we vectorized the words using a static Word2Vec dictionary obtained by training a model on a large Turkish corpus containing over 400,000 unique words.[8] For each sentence, we applied a Bidirectional GRU, or Bi-GRU, to generate contextualized word representations by considering both past and future words, i.e., tokens, within the sentence. Next, we applied a word-level self-attention over the Bi-GRU outputs to obtain sentence embeddings. These sentence embeddings were then passed through another Bi-GRU to capture the context of each sentence within the script. Finally, we applied sentence-level self-attention to generate a single embedding that represents the entire script. Fig. 3 illustrates the HAN model, where S_i refers to the ith sentence in a script and w_j^k refers to the jth word in the kth sentence.

3.4. BigBird model

BigBird is a BERT-like model that employs a sparse attention mechanism, allowing it to scale linearly with document length, unlike the full attention mechanism used in standard BERT, which scales quadratically with the length [16]. This makes it feasible to apply BERT-like models to long documents—up to 4096 tokens. Since the scripts of Turkish TV series are often similar in length to movie scripts, typically spanning thousands of tokens, BigBird appears to be a promising model for our task.

In the sparse attention mechanism in BigBird, tokens attend to their nearby neighbors (local/window attention), and to a few randomly selected tokens (random attention). In addition, certain special tokens — most commonly the [CLS] token — are allowed to attend to all other tokens in the input, which is referred to as global attention. Fig. 4 illustrates this attention pattern.

The standard way of using BigBird for a downstream task is similar to using a regular BERT model. Typically, the [CLS] token's output from the final hidden layer of a pretrained BigBird model is taken as the document embedding, and a classification head is added on top. However, since no pretrained BigBird model exists for the Turkish language, we pretrained a custom BigBird model from scratch using the

[7] https://huggingface.co/dbmdz/bert-base-turkish-cased.

[8] https://github.com/akoksal/Turkish-Word2Vec.

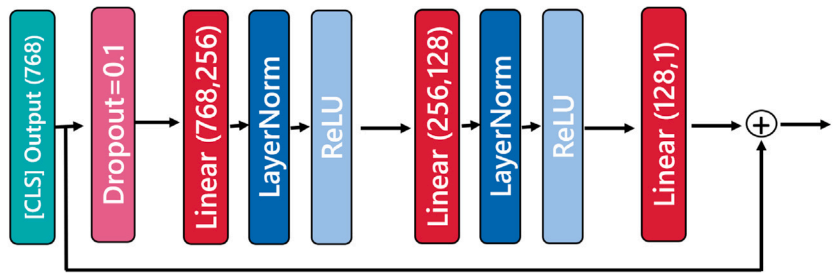


Fig. 2. Classification head of the truncated BERT model.

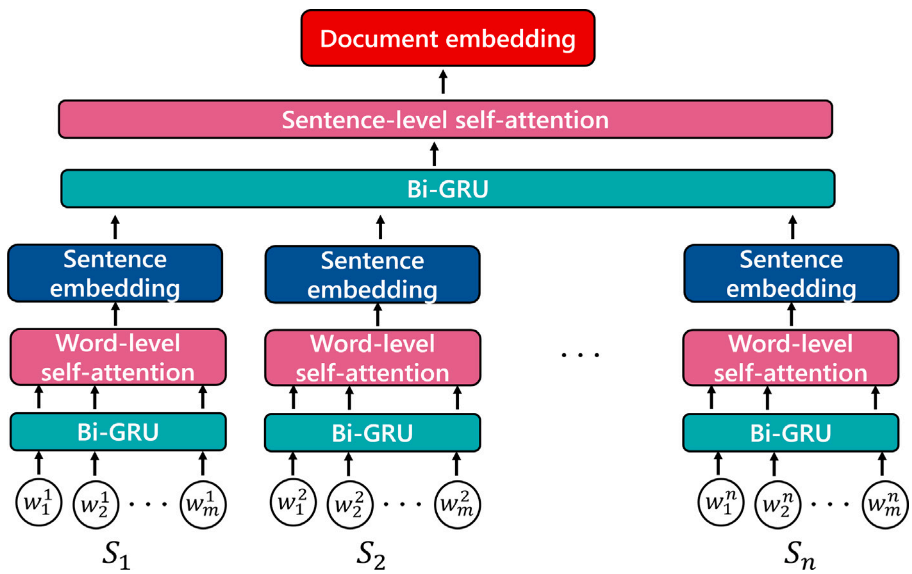


Fig. 3. Hierarchical attention network model.

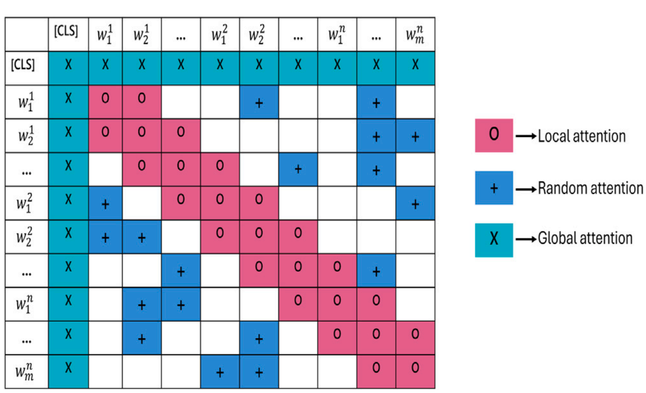


Fig. 4. The sparse attention mechanism in BigBird.

Turkish Wikipedia corpus. For this task, we used an ml.g4dn.12xlarge instance on Amazon SageMaker, which comes with 4 NVIDIA Tesla T4 GPUs (16 GB each), 48 vCPUs, and 192 GiB of RAM. We trained the model using the Hugging Face Transformers library, enabling the block-sparse attention mechanism with a block size of 64 and a maximum input length of 4096 tokens. Our model includes 4 transformer layers, each with 11 attention heads and a hidden size of 220.[9] Under this setting and using the instance of Amazon SageMaker described above, we pre-trained a BigBird model from scratch on a Turkish Wikipedia corpus consisting of approximately 2.5 million sentences and 30 million tokens.[10]

Finally, we used the output of the [CLS] token from the pretrained BigBird model as the document embedding and added a classification head identical to the one used with the truncated BERT model, as shown in Fig. 2.

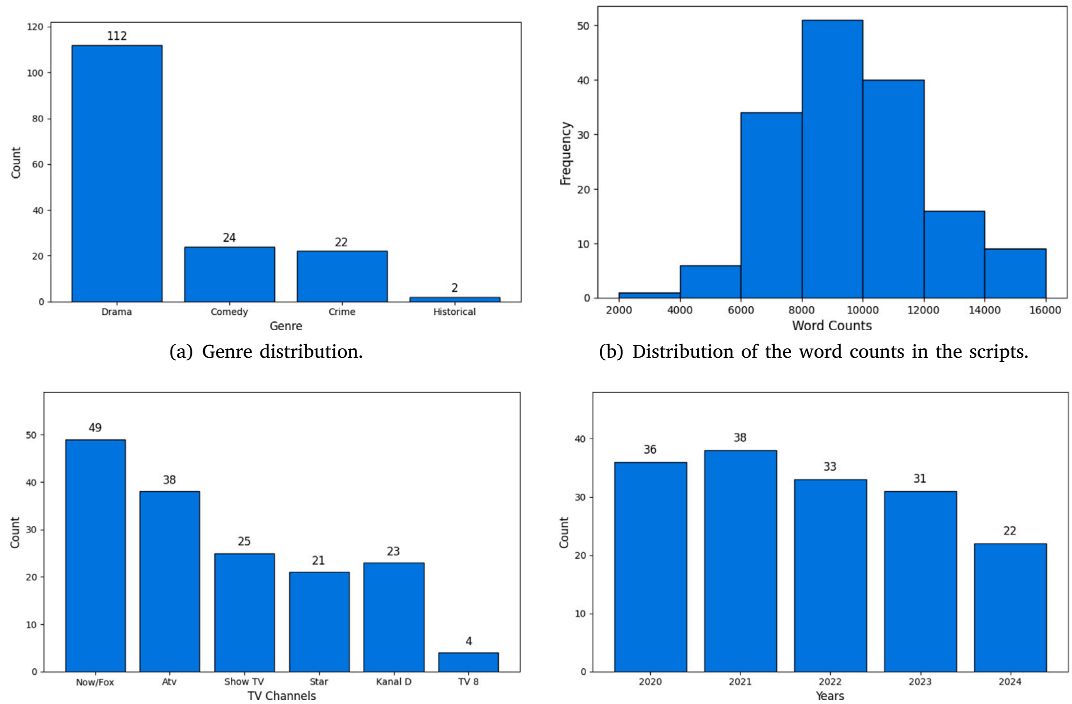
Continual domain-adaptive pretraining

As will be explained in the next section, our labeled dataset is relatively small, which may lead to overfitting during training. To mitigate this issue and better align the base BERT models used in the Truncated BERT and Turkish BigBird models with the domain of Turkish TV series scripts, we performed an additional pretraining step. This process, commonly referred to as *continual domain-adaptive pretraining* in the literature, involves training base models on unlabeled domain-specific corpora with the goal of improving end-task performance [30].

Concretely, we created an additional dataset consisting of the first episodes of approximately 600 Turkish TV series aired between 2009 and 2019, ensuring that the scripts in this dataset were not included in the original dataset used for training and testing. Using this dataset, we further pre-trained the base models mentioned above via masked language modeling (MLM). Specifically, we followed the standard MLM procedure, where a portion of the tokens (typically 15%) in each sentence are randomly masked and the model is trained to predict

[9] Full configuration can be easily loaded using the Transformers library: `AutoConfig.from_pretrained ("FiratIsmailoglu/bigbird-turkish-pretrained")`.

[10] https://corpora.wortschatz-leipzig.de/en?corpusId=tur_wikipedia_2021.



(a) Genre distribution.

(b) Distribution of the word counts in the scripts.

(c) TV channel distribution.

(d) Year distribution.

Fig. 5. Descriptive statistics of the scripts in the dataset.

these masked tokens. Finally, we employed these further pre-trained base models as encoders and added classification heads on top of them within both the truncated BERT and BigBird architectures.

4. Experiments

4.1. Dataset

For this project, we collected the scripts of the first episode of 160 TV series that aired on Turkey's major mainstream channels and completed in the last five years. Fig. 5 presents general statistics about the collected TV series scripts. As shown in the figure, drama emerges as the dominant genre, accounting for approximately 70% of the entire dataset. The number of words per script varies between 2000 and 10,000, with a roughly normal distribution centered around a mean of 8000 words. Additionally, because some channels, such as Now (formerly Fox), produce more series than others, like TV8, the scripts are not evenly distributed across TV channels. Finally, since the volume of TV series production varies from year to year, the scripts in our dataset are also unevenly distributed across the years 2020 to 2024.

The total number of episodes for these series ranges from 3 to 177. Around 42% of the series lasted fewer than 13 episodes (a common threshold in the Turkish media industry) and are considered unsuccessful thus are labeled as Class 0. The remaining 58% are labeled as Class 1 and considered successful. Fig. 6 shows labels statistics of the dataset.

Data augmentation to enhance the dataset

As in the case of the other deep learning architectures, LLMs also require large amounts of training data. To increase the amount of data and introduce more diversity among training examples, we applied several *data augmentation* techniques [31]. First, as a paraphrasing-based method, we used back-translation: the original Turkish scripts were translated into English and then back into Turkish using the Google Translate API. Second, we applied two noising-based augmentation techniques—one at the word level and one at the sentence level. For word-level augmentation, we used synonym replacement, where each word in the script was replaced with one of its synonyms with a probability of 0.3. This was done using a Turkish synonym dictionary[11] containing over 34,000 words. For sentence-level augmentation, we randomly shuffled the order of sentences within a script. Note that all data augmentation was applied only to the training scripts, while the test scripts were kept unchanged.

4.2. Evaluation criteria

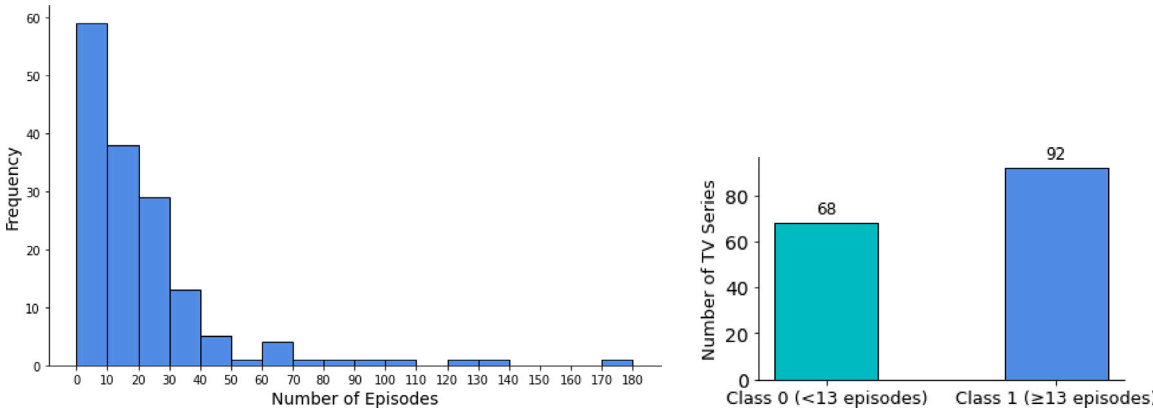
In this research, our main goal is to predict the success of Turkish TV series. We approach this through two subtasks: (1) predicting the total number of episodes a series will have, and (2) predicting whether a series will be successful or not, defined as having fewer than 13 episodes. The first subtask is formulated as a regression problem, so we evaluate model performance using Mean Squared Error (MSE). The second subtask is a binary classification task, and we assess model success using confusion matrices along with the metrics Accuracy, True Positive Rate (TPR) and True Negative Rate (TNR) derived from them. For reference, these metrics are defined in Fig. 7.

Here n denotes the number of training/testing instances, and Y_i (resp. $\hat{Y}_i$) represents the true (resp., predicted) total number of episodes for the ith TV series. TP refers to true positives, and TPR (True Positive Rate) measures how well the model identifies successful series (Class 1). TN refers to true negatives, and TNR (True Negative Rate) reflects the model's ability to correctly identify unsuccessful series (Class 0).

4.3. Experimental results

To ensure a fair evaluation and to assess whether the observed differences between models are statistically significant, we conducted 10-fold cross-validation and applied a paired t-test. In each iteration,

[11] https://huggingface.co/datasets/agmmnn/turkish-thesaurus-synonyms-antonyms.



(a) Frequency of number of episodes of the TV series. (b) Class distribution of 0 and 1 classes.

Fig. 6. Label statistics of the dataset.

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

		Predicted: Class 0	Predicted: Class 1
Actual	Class 0	TN	FP
Actual	Class 1	FN	TP

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \qquad \mathrm{TNR} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}}$$

$$\mathrm{ACC} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}}$$

Fig. 7. Evaluation metrics considered in this study.

Table 1
The MSE scores of the models.

Model	MSE
TF-IDF-Lasso	517
TF-IDF-Ridge	604
Truncated BERT	514
HAN	**485***
BigBird	507

Table 2
Performance scores of the created NLP Models.

(a) TPR scores		(b) TNR scores		(c) Accuracy scores	
Model	TPR	Model	TNR	Model	ACC
TF-IDF-LR	0.41	TF-IDF-LR	0.84	TF-IDF-LR	0.63
TF-IDF-RF	0.39	TF-IDF-RF	0.82	TF-IDF-RF	0.61
Truncated BERT	0.46	Truncated BERT	0.79	Truncated BERT	0.62
HAN	**0.71***	HAN	0.82	**HAN**	**0.77***
BigBird	0.61	**BigBird**	**0.86**	BigBird	0.73

we used around 90% percent of data as training set and the remaining data as test set. Then, we tripled the size of the training set by applying data augmentation methods: back-translation, synonym replacement, and sentence shuffling. The results reported below represent the average performance on the original (non-augmented) scripts from the test set of each cross-validation fold. Statistically significant differences between the best and second-best performing models, determined using a paired t-test ($p < 0.05$), are marked with an asterisk (*).

4.3.1. Regression results

The MSE scores that each of the models achieved are given in Table 1.

As shown in Table 1, HAN performs best among the regression models. Based on the paired t-test conducted over the MSE scores from each fold of the 10-fold cross-validation, the HAN model performs significantly better than the second-best model, namely BigBird. However, unlike the TF-IDF-based regression models (i.e., TF-IDF-Lasso and TF-IDF-Ridge), the BERT-based models tend to produce predictions with very low variance. Specifically, we observed that the truncated BERT, HAN, and BigBird models frequently predict values close to the average episode count (around 16) for the majority of input scripts. In other words, these models exhibit a tendency to collapse toward the mean of the target values. This behavior can be largely attributed to the limited size of the training set, which fails to capture sufficient variation across scripts. Nevertheless, this issue can be partially mitigated by applying early stopping during training, albeit at the cost of slightly increased MSE scores.

4.3.2. Classification results

We show the average TPR, TNR, and accuracy scores for each model, computed over 10-fold cross-validation in Table 2.

As shown in Table 2, the HAN model demonstrates the strongest performance for the binary classification of the scripts, consistent with its results in the regression task. It achieves the highest accuracy and excels at distinguishing between successful and unsuccessful scripts, as reflected in its high True Positive Rate (TPR) and True Negative Rate (TNR) scores. We also found that its performance in terms of TPR and accuracy is statistically superior to that of the other models.

The BigBird model also exhibits robust performance and can be considered a reliable alternative for this task. It achieves the highest TNR score, although this difference was not found to be statistically significant. Notably, there is a clear performance gap between these two models and the remaining ones. The other models perform slightly better than random guessing. In fact, they tend to be biased toward predicting Class 0, frequently outputting negative labels, which is reflected in their low TPR scores and correspondingly high TNR scores. This tendency also applies to a lesser extent to the BigBird model.

Overall, these results suggest that the transformer-based large language models (LLMs) generally outperform TF-IDF-based models, indicating the necessity of LLMs for predicting the success of TV series based on their first episode scripts. However, the truncated BERT model stands as an exception. Despite being a BERT-based model, it performs worse than both the TF-IDF-based baselines in general. The likely explanation lies in its inability to process long sequences effectively,

Table 3
Predictions made by the BigBird and HAN models for the Iconic Turkish TV Series.

	Drama	Comedy	Crime	Historical
Successful	4	2	4	2
Failure	2	5	0	1
Total	6	7	4	3

as it truncates nearly %90 of the script. This limitation makes it the least effective option among all the models we tested.

Despite the HAN model being the best-performing among all models evaluated, its performance may be underappreciated at first glance, particularly since its accuracy does not reach the %90 range that is often expected in many classification tasks. While this observation is valid to some extent, it is important to emphasize that the predictions are made solely based on the scripts of the first episodes. In reality, the success of a TV series can depend on a variety of external factors—such as cast popularity, broadcasting channel, airing schedule, and even seasonal timing (e.g., summer vs. winter programming). In this study, we make a deliberate simplification by ignoring these external variables and relying exclusively on script content. As a result, if the script is not the main reason behind a show's success, the predictive power of the models is expected to be limited.

Nonetheless, our findings highlight that the two transformer-based large language models (LLMs) we developed, i.e., the HAN and BigBird models, are effective at distinguishing promising scripts from less favorable ones. From this perspective, LLMs demonstrate potential in identifying high-quality content at the pre-production stage. Such predictive tools could help producers get a sense of a script's potential before investing a lot of money into production.

4.4. Evolution of audience preferences

Remember that our models were trained on a dataset of recent Turkish TV series, meaning their predictions are shaped by current viewer trends. As such, when we apply our models to older series, we are essentially judging past shows based on today's standards, which can help us identify whether audience tastes have changed over time. To explore this, we selected 20 iconic Turkish TV series that aired over five years ago meaning that they are not part of our training set. Also, each of these series lasted more than 13 episodes, has an IMDb rating above 8 and falls into one of four main genres: drama, comedy, crime, or historical.

In the experiments above, the BigBird and HAN models emerged as the most successful. We used both to evaluate the older TV series. In cases where these two models disagreed, i.e., one predicting success and the other failure, we relied on the prediction from the TF-IDF-LR model, which was the third-best performing model. The final ensemble results are presented in Table 3.

Assuming that the combined judgment of the BigBird and HAN models reflects today's TV standards, we can claim that the preferences of Turkish TV audiences have not changed dramatically over the years. Most of the iconic series in our sample would likely still be considered engaging if they aired today. However, it is worth noting that 5 out of 7 classical comedy series were predicted to be unsuccessful by today's standards, suggesting a possible shift in audience tastes when it comes to comedy. This might indicate that the sense of humor in Turkey have evolved over time.

5. Limitations

We also find it important to acknowledge the limitations of our study. First and foremost, the models and experiments presented here are language-specific. Since both the training data and the pretrained language models used in our work are exclusively in Turkish, the developed predictive models are inherently limited to the Turkish language. As a result, these models are unlikely to generalize well to TV series scripts written in other languages such as English or Spanish.

Another major limitation is the size of the labeled dataset. Although we attempted to mitigate this by applying continual domain-adaptive pretraining through masked language modeling (MLM) using scripts from 600 Turkish TV series, the final fine-tuning was performed on a much smaller set—approximately 160 scripts. This limited size likely constrained the full capacity and generalization power of the final models.

Lastly, we note that our BigBird-based models are restricted to processing at most 4096 tokens, which is a limitation of the architecture itself. Consequently, only the first 4096 tokens of each script were considered during training and testing.

6. Conclusion

With the recent breakthroughs in Large Language Models (LLMs), these models have found success in a variety of applications and have become an integral part of our daily lives. LLMs have also begun to shape the media and entertainment industry, particularly in the area of content creation. However, their predictive potential in this area has not yet been fully explored, leaving plenty of room for further exploration. In this study, we suggest that it is possible to estimate audience interest in a TV series based solely on its script using LLMs before the show is created and before producers make an investment.

In this study, we focus on Turkish TV series, as Turkey ranks as the third largest exporter of TV series globally, reaching around 800 million viewers worldwide. Also, these series are produced in Turkish, a low-resource language, which makes our research both valuable and necessary. By creating a dataset consisting of the scripts of the first episode of Turkish TV series aired in the last five years, we aim to train LLMs to predict the success of a series based on its script alone. Here, one of the main challenges we encountered is the length of these scripts, as each episode is usually around two hours long, resulting in scripts that average around 8000 tokens—far exceeding the 512-token input limit of standard BERT-based language models.

To address the challenge of processing long Turkish texts, we pre-trained a BigBird model — a sparse variant of BERT designed for long sequences — for Turkish, which is the first example of such work in the literature. We then fine-tuned this model for the task of predicting TV series success. In addition, we developed a Hierarchical Attention Network (HAN) model, which can also support long document inputs. To establish a broader benchmark and highlight the importance of LLMs, we additionally built two TF-IDF-based models using the same training data. We found that the BigBird and HAN models performed best in predicting the total number of episodes based on the first episode scripts, as indicated by their relatively low MSE scores. However, both models tended to generate predictions close to the average episode count across most scripts.

We also framed the task as a binary classification problem, labeling TV series that did not reach 13 episodes as unsuccessful (negative) and the others as successful (positive). This threshold aligns with a common practice in the Turkish media industry, where initial contracts for TV series are typically signed for 13 episodes, so fewer episodes often signal early termination due to low performance. Experimental results showed that the HAN model achieved the best performance, reaching an accuracy of 0.77 in average along with high true positive and true negative rates. The BigBird model, another LLM tailored for long texts, followed with an accuracy of 0.73. The TF-IDF methods the truncated BERT model were slightly better than random guessing.

While the results might not seem impressive at first glance since they do not reach an accuracy of around 90%, we still find them quite promising. It is worth noting that our predictions are based only on the scripts of the first episodes, assuming that these episodes reflect the full potential of the series. Of course, when this assumption does not hold, even the best LLMs can fall short. Thus, we respond positively to our research question: when trained on a sufficiently large corpus of TV scripts a well-designed and properly trained LLM can indeed uncover the potential success of a series before it is even produced.

As a side task, we also explored whether audience preferences in Turkey have shifted over time by leveraging our best performing LLMs, i.e., HAN and BigBird, trained on recent TV series. To do this, we applied these models to several iconic Turkish TV series from various genres that were highly popular in the past. Assuming these models reflect today's standards for TV content, we found no significant change in audience preferences, as most of the past series were still classified as successful. The notable exception was in the comedy genre: a majority of the older comedy series were predicted as unsuccessful by our models, suggesting that the sense of humor among Turkish audiences may have evolved over the years.

Since traditional BERT models scale quadratically with the length of the input, classifying long documents remains an active research area, with new models and techniques emerging regularly. Future work will focus on adapting these advancements to the task of predicting the success of TV series. In addition, we plan to expand our research in terms of model interpretability aiming to reveal the key concepts and themes within scripts that contribute to a series being perceived as successful or not.

Declaration of Generative AI and AI-assisted technologies in the writing process

The author used ChatGPT to improve readability and check grammar, but the outputs of this tool were not used as is, instead they were reviewed and adjusted by the author before being included.

Declaration of competing interest

The author declares that there is no conflict of interest regarding the publication of this manuscript.

Data/model availability

Data will be made available on request. The created models can be run at:
https://huggingface.co/spaces/FiratIsmailoglu/TurkishDramaPredictor.

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